FU Xiao

Tel:(+86)19858125696 | Email: lemonaddie0909@zju.edu.cn | Homepage: https://fuxiao0719.github.io/|

Education

Zhejiang University Hangzhou, Zhejiang
Bachelor in Information Engineering | College of Information Science & Electronic Engineering *Sept. 2018 – Jul. 2022*

- **GPA:** 3.96/4.00, 90.45/100 (92.33/100 for 2^{nd} year), **Ranking:** 6/146
- **National Scholarship awardee, Twice**
- **Chu Kochen Honors College:** Intensive Training Program of Innovation and Entrepreneurship (ITP) (Honored minor for the selected top 40 students with innovation capability from 6,000 freshmen)

Publications

∗ indicates equal contribution

- [C1] Guangyi Zhang*, **Xiao Fu***, Qiyu Hu, Yunlong Cai, Guanding Yu, "Hybrid Precoding Design Based on Dual-Layer Deep-Unfolding Neural Network," *2021 IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Helsinki, Finland, 2021, pp. 678-683 **[Paper][Code]** **(Two "Strong accept" out of three random reviewers)**

Research Experience

Dynamic Perceptive Network for Monocular Depth Estimation Jul. 2021 – Sept. 2021
Supervisor: Prof. *Xiaojuan Qi* *CVMI Lab, The University of Hong Kong*

- Propose a neural network based on dynamic routing framework, which could generate adaptive receptive field size in response to arbitrary resolution of input images, tackling the problem of the inconsistent structure in high-resolution monocular depth estimation
- For one high-resolution(1443×1082 for IBMS-1) depth estimation on GPU Tesla V10, OURS consumes 1.03s compared to 11.24s for SoTA, which means our model requires less computational complexity
- As a tradeoff for less complexity and faster deducement, the performance of OURS drops around 4% on ordinal error(ORD) and RMSE metrics, tested on datasets:Middleburry 2014 and IBMS-1

Machine Learning for Wireless Network Complexity Optimization Apr. 2020 – Dec. 2020
Supervisor: Prof. *Guanding Yu* *Zhejiang University*

- Proposed a novel framework, i.e. a Dual-Layer Deep-Unfolding Neural Network(DLDUNN) which unfolds the iterative Penality Dual Decomposition (PDD) algorithm into a layer-wise structure whereby some trainable parameters are introduced into the places of the high-complexity operations, e.g. large-dimensional matrix inversions and iterative sub algorithms, so as to address the high computational complexity problem of the dual-layer PDD algorithm and presents the major original performance in the meantime
- **Laureate of Excellent Award(Team leader) - Student Research Training Program(Province-level)**

Selected Honors and Scholarships [Certificate & Report]

- National Scholarship **(Highest Honor for Chinese Undergraduates, Top 1.8%)** *2020,2021*
- ISEE Yuelun Alumni Scholarship **(6/1,150 selection ratio, Bonus 20,000 CNY)** *2020*
- First-Class Scholarship for Academic Excellence of ZJU & National Talents Training Base **(Top 3%)** *2020,2021*
- Government Scholarship for Academic Excellence of Zhejiang Province **(Top 5%)** *2019*
- Finalist Winner – Mathematical Contest in Modeling (MCM/ICM) **(Top 2% out of 20,954 teams)** *2020*
- Meritorious Winner – Mathematical Contest in Modeling (MCM/ICM) **(Top 7% out of 26,112 teams)** *2021*
- Gold Prize – 7^{th} International "Internet+" Innovation and Entrepreneurship Contest **(Team Leader)** *2021*
- Gold Prize – 12^{th} "Challenge Cup" National College Student Business Plan Competition *2021*
- Grand Prize – "Yongdian Cup" Innovation and Entrepreneurship Contest **(Team Leader)** *2019*
- Academic Excellence Scholarship of Zhejiang University *2019,2020,2021*
- Zhejiang University Five-Star Volunteer *2021*